Swinepox

Chi-Fei Kao (高啓霏)

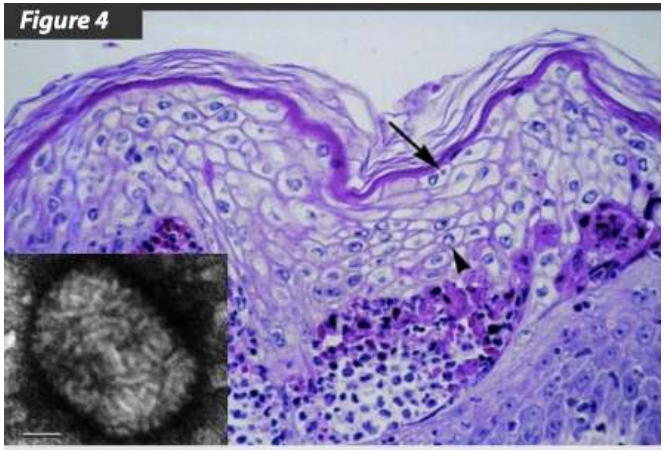
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Introduction

- An acute, mild and usually self-limiting skin disease in pigs
- Characterized by pustular (pox) skin lesions in localized areas, with more severe forms in young pigs (<3–4months).
- Worldwide, usually associated with **poor sanitary conditions**



Skin; Pig 7: generalized distribution of cutaneous pox lesions in a 1-day-old piglet



Skin; Pig 7: hyperplastic epidermis showing intraepidermal pustules, ballooning degeneration, eosinophilic intracytoplasmic inclusions (arrow), and intranuclear vacuoles (arrow head); HPS stain; bar = 30 µm

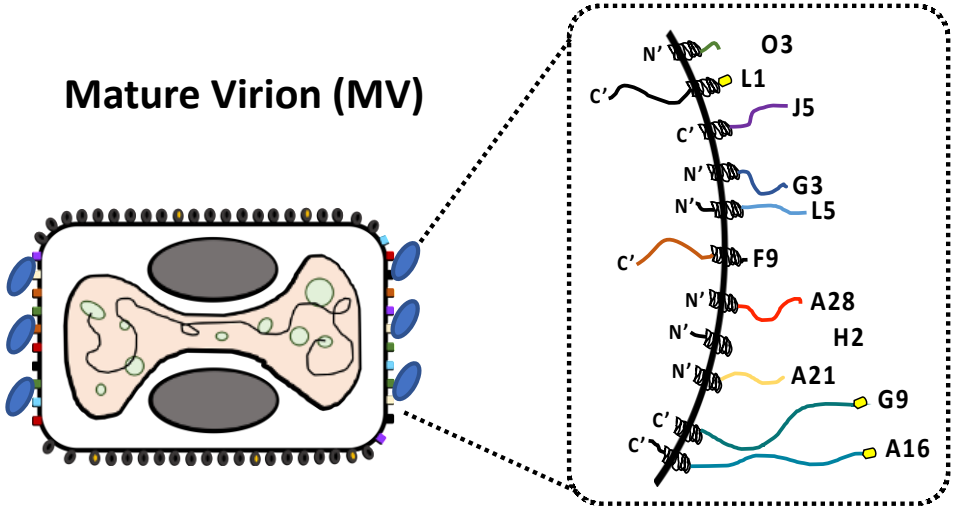
Etiology

Swinepox virus (SwPV)

- Genus Suipox*virus*, subfamily of *Chordopoxvirinae*
- Large **brick-shaped**, **enveloped**, & approximately 240-310 nm
- Linear double-stranded DNA genome of 146 kb comprising a conserved central coding region with inverted terminal repeat sequences at both ends
- Two infectious particles: mature virion (MV) & extracellular virion (EV)
- SwPV antigenically distinct from other poxviruses.
- Genetically stable

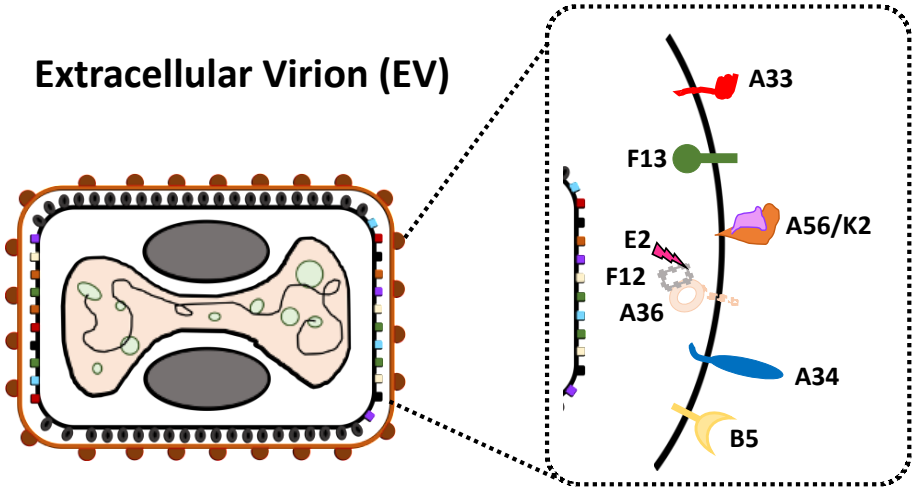
Two Infectious Forms of Viral Particles

Mature Virion (MV)



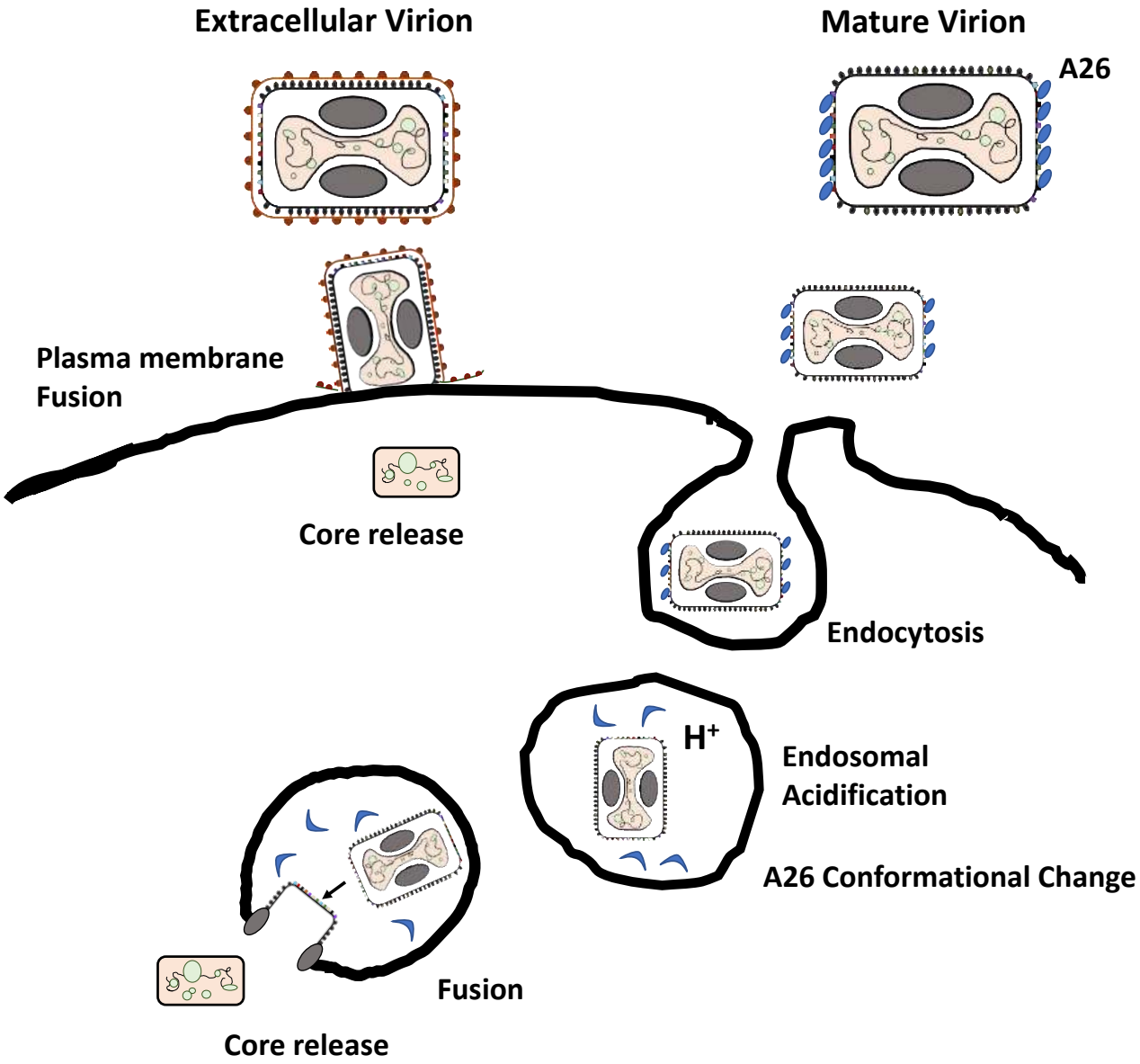
- **A26⁺**
- **Environmental Resistant, >95%**
- **One layer of membrane derived from ER**
- **Release upon cell lysis**

Extracellular Virion (EV)



- **A26⁻**
- **Fragile, <5%**
- **Additional layer of membrane derived from Golgi**
- **Cell-cell membrane fusion for viral spreading**

Two Entry Pathways of MV & EV



Epidemiology

- First reported in North America in 1929
- Pig is the only natural host of SwPV

  → **No zoonotic potential**

- SwPV failed to cause lesions in the scarified skin of rabbits and guinea pigs nor infected kidney cells from cattle and rabbits
- Historically vaccinia virus was also reported relevant to swine pox diseases (pustular skin lesions) due to the eradication of smallpox in human; now SwPV appears to be the only cause.

Epidemiology

- Transmission
  - ✓Most commonly between swine by the bite of the **pig louse**, ***Haematopinus suis***
  - ✓Flies (*Stomoxys calcitrans*) and mosquitoes may also be mechanical vectors for SwPV transmission
  - ✓Direct contact
  - ✓**Vertical transmission**





By Daniel J. Drew - Gall L (2019). Invertebrate Zoology Division, Yale Peabody Museum. Yale University Peabody Museum. Occurrence dataset https://doi.org/10.15468/0lkr3w accessed via GBIF.org on 2019-07-07. https://www.gbif.org/occurrence/1576192043, CC0, https://commons.wikimedia.org/w/index.php?curid=80233842

Pathogenesis

- SwPV enters the host **through a break in the skin** and then replicates in the cytoplasm of **keratinocytes** of the stratum spinosum.
- Route of viremia remains unclear.
- It has been suggested that congenital infections may result from viremic sows, but virus has not been isolated from blood samples from infected animals.

Clinical Presentation

The clinical signs of SwPV infection are **age-dependent**.

- Clinical sign:
  - ✓ Maculae, papulae, and vesicles to umbilicated lesions with
    斑疹　　丘疹　　水泡
    pustular content, followed by crusting
    膿胞　　痂皮
  - ✓ ± Slight elevation of body temperature and loss of appetite in young pigs
  - ✓ Cutaneous lesions on the flanks, ventral abdomen, legs, inguinal areas, and ears and less frequently on the face near the eyes (adult)
  - ✓ In congenital SwPV infections, lesions may be observed over the entire body



Skin Lesions–Adult Pigs













Kumar et al., 2023

Skin Lesions–Neonatal Piglets



The skin of the piglet is covered by erythematous macules with an ulcerated center.

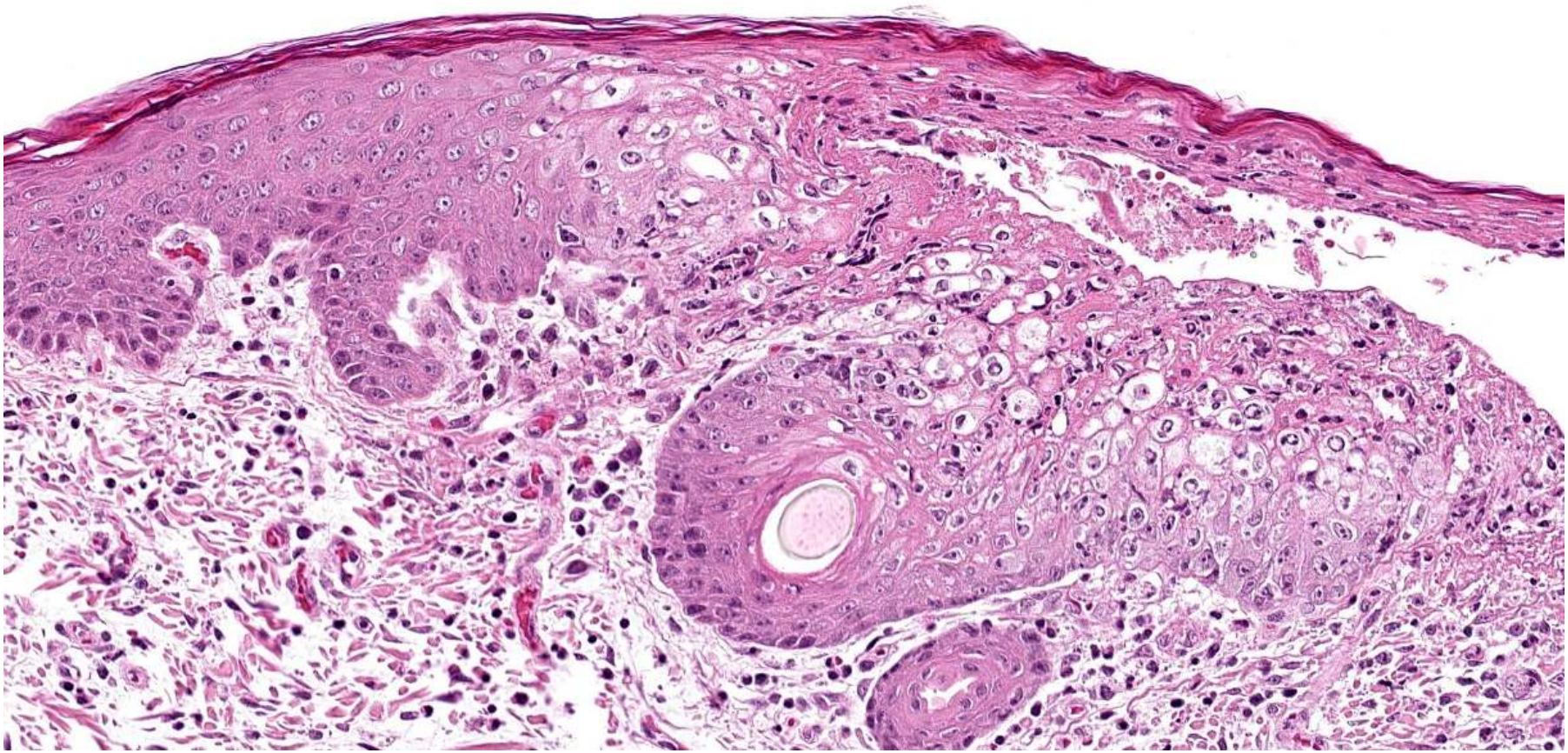
https://www.askjpc.org/wsco/wsc_showcase2.php?id=bURpU3JmN2VrV0k3Z0pYZzlPOGpvZz09

Clinical Presentation

- Morbidity may be high but mortality is low
- Secondary dermatitis may accompany the virus-induced alterations
- **Stillbirth or neonatal death** within a few days in congenital infection; the sow, however, may appear clinically normal.

Lesions

- The classic histological alteration of SwPV infection is the **hydropic degeneration** of the stratum spinosum **keratinocytes** of the epidermis, occurring in the papular stage of lesions.
- **Eosinophilic Intracytoplasmic inclusion bodies** in the swollen infected cells with a vacuolated nucleus can also be found.
- The rupture and coalescence of epidermal cells may form a few small **vesicles**. Subsequent **inflammatory infiltration** into the underlying dermis converts vesicles to pustule.



There are necrotic and degenerating keratinocytes, are infiltrated by low to moderate numbers of necrotic neutrophils admixed with cellular debris.

https://www.askjpc.org/wsco/wsc_showcase2.php?id=bURpU3JmN2VrV0k3Z0pYZzlPOGpvZz09



Swollen keratinocytes contain one or multiple brightly **eosinophilic intracytoplasmic inclusion bodies** (arrows)

https://www.askjpc.org/wsco/wsc_showcase2.php?id=bURpU3JmN2VrV0k3Z0pYZzlPOGpvZz09

Laboratory Diagnosis

A presumptive diagnosis of swinepox is based on the observation of typical pox lesions on the skin of affected animals, especially under poor sanitary conditions.

- Differential diagnosis:
  - ✓ **Vaccinia virus**
  - ✓ **Vesicular diseases**: FMD, SVD, SVV, VES & VS
  - ✓ Skin lesions associated with classical swine fever virus, erysipelas, pityriasis rosea, streptococcal dermatitis, vegetative dermatitis, parasitic skin disease, allergic skin lesion, nutritional disorders and sunburn.

Laboratory Diagnosis

EM and histopath may confirm involvement of SwPV.

- Virus detection
  - ✓ Identification of (infectious) virion: EM, histo, virus isolation

    *using primary porcine kidney cells or PK15 cell line; blind passages
  - ✓ Identification of nucleic acids: PCR, qPCR
- Serological test
  - ✓ ELISA
  - ✓ Precipitation antibody tests
  - ✓ Immunoelectroosmophoresis (IEOP)
  - ✓ Virus neutralization: Swine do not develop high levels of neutralizing antibody, therefore, negative neutralizing antibody results should be interpreted with caution.

Prevention & Control

- There is no specific treatment for swinepox.
- Antibiotic treatment is recommended if it is necessary to control secondary bacterial infections.
- Swinepox has a negligible economic impact, and no vaccine has been developed.
- The strict host-restriction has enabled SwPV as a promising viral vector for vaccine development.
- The best prevention and control is good animal husbandry, including ectoparasite control.

下列有關豬痘之敘述,何者正確?

(A)成豬病變較仔豬嚴重
(B)豬是此病致病原之唯一宿主
(C)在豬群中之發病率及死亡率都高
(D)臺灣為此病之清淨地區

下列有關豬痘病毒致病機轉之敘述，何者錯誤？

(A) 經由皮膚損傷處感染宿主
(B) 可在表皮之角質細胞（keratinocyte）增殖
(C) 蝨子可機械性傳播本病
(D) 成豬較易感染

有關豬痘感染豬隻後之症狀及病變，下列敘述何者正確？

(A) 以多發性皮膚病變呈現
(B) 常集中於臉部皮膚
(C) 會有高燒不退之症狀
(D) 可在皮膚病變區發現嗜鹼性質內包涵體

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下列有關豬痘之敘述，何者錯誤？

(A)豬為豬痘病毒的唯一宿主
(B)仔豬較成豬容易感染
(C)發病率及死亡率均高
(D)豬蝨為機械性傳播媒介